

SOLIDWORKS API (.NET VBA C#) — Reference

(291/337): ISMNormalCutFeatureData2- ISplineHandle

SOLIDWORKS API Documentation (converted from CHM)

ISMNormalCutFeatureData2 (Type)

SOLIDWORKS API Help

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ISMNormalCutFeatureData2 Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISMNormalCutFeatureData2 Interface

Allows access to a sheet metal normal cut feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISMNormalCutFeatureData2
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2
```

C#

```
public interface ISMNormalCutFeatureData2
```

C++/CLI

```
public interface class ISMNormalCutFeatureData2
```

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2](#).

Example

'VBA

- '1. Open **c:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2019\samples\whatsnew\sheet metal\normal_cut.sldprt**.
- '2. Open the Immediate window.
- '3. Run the macro.
- '4. Creates **Normal Cut1** in the FeatureManager design tree.
- '5. Edits the feature by specifying a normal cut direction entity.
- '6. Inspect the Immediate window.

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swFeatMgr As SldWorks.FeatureManager
Dim swFeat As SldWorks.Feature
Dim swNormalCutFeatData As SldWorks.SMNormalCutFeatureData2
Dim grpData As SldWorks.SMNormalCutGroupData
Dim swSelMgr As SldWorks.SelectionMgr
Dim swCutDirectionFace As SldWorks.Face2
Dim errCode As Long
Dim boolstatus As Boolean
Dim varGrpNames As Variant
Dim i As Long
Dim j As Long
Dim Name As String
Dim Face As SldWorks.Face2
```

Sub main()

```
Set swApp = Application.SldWorks
Set swModel = swApp.ActiveDoc
Set swModelDocExt = swModel.Extension
Set swFeatMgr = swModel.FeatureManager
Set swNormalCutFeatData = swFeatMgr.CreateDefinition(swFmNormalCut)
```

```

swNormalCutFeatData.AutoPropagation = True
swNormalCutFeatData.OptimizeGeometry = True
swNormalCutFeatData.NormalCutParameters = swNormalCutExtent

' Select the face(s) to make normal

boolstatus = swModel.Extension.SelectByRay(4.71296104838217E-02, 1.42054003862171E-02, -6.68923799798904E-03,
-0.13322686545804, 0.182153484913837, 0.974202602261958, 6.19874904830013E-04, 2, True, 1, 0)

Set grpData = swNormalCutFeatData.CreateGroup(errCode)

' Create the normal cut feature

Set swFeat = swFeatMgr.CreateFeature(swNormalCutFeatData)

' Modify the normal cut feature

Set swNormalCutFeatData = swFeat.GetDefinition()
swNormalCutFeatData.AccessSelections swModel, Nothing
varGrpNames = swNormalCutFeatData.GetGroupNames()

For i = 0 To UBound(varGrpNames)
    Name = varGrpNames(i)
    Debug.Print "GroupName: " & Name
    Set grpData = swNormalCutFeatData.GetGroupByName(Name)
    Dim varFaces As Variant
    varFaces = grpData.Faces
    For j = 0 To UBound(varFaces)
        Set Face = varFaces(j)
        Debug.Print "Face " & j & " area: " & Face.GetArea
    Next j
    Debug.Print "-----"
Next i

'Specify the cut direction

Set swSelMgr = swModel.SelectionManager
boolstatus = swModel.Extension.SelectByRay(-0.1499999999999892, 2.58556514245498E-02, -4.63505465353364E-03,
0.393242668518252, 0.12460738379391, -0.910951811876282, 1.22626958447854E-03, 2, False, 0, 0)
Set swCutDirectionFace = Nothing
Set swCutDirectionFace = swSelMgr.GetSelectedObject6(1, -1)

Dim swSelectEnt As SldWorks.Entity
Set swSelectEnt = swCutDirectionFace
boolstatus = swSelectEnt.Select4(False, Nothing)
swNormalCutFeatData.CutDirection = swCutDirectionFace

Dim isModified As Boolean
isModified = swFeat.ModifyDefinition(swNormalCutFeatData, swModel, Nothing)

End Sub

```

Example

[Create Normal Cut Feature \(C#\)](#)

Remarks

Extruded cuts that pierce models at an angle have non-normal side walls. To make these side walls normal, you create a normal cut feature.

To create a normal cut feature:

1. Open a sheet metal part that has a cut-extrude with non-normal side walls.
2. Call `IFeatureManager::CreateDefinition(swFmNormalCut)` to access this interface.
3. Set non-entity properties on this interface.
4. Use `IModelDocExtension::SelectByRay` or `IModelDocExtension::SelectById2` with Mark = 4 to select the offset plane, only if `ISMNormalCutFeatureData2::NormalCutParameters` is set to `swNormalCutParameters_e.swNormalCutOffsetPlane`.
5. Use `IModelDocExtension::SelectByRay` or `IModelDocExtension::SelectById2` with Mark = 8 to select the direction of the normal cut, only if `ISMNormalCutFeatureData2::NormalCutParameters` is set to `swNormalCutParameters_e.swNormalCutExtent`.
6. Use `IModelDocExtension::SelectByRay` or `IModelDocExtension::SelectById2` with Mark = 1 to select each non-normal face in the group. If `ISMNormalCutFeatureData2::AutoPropagation` is set to true, then all faces tangent to a selected face are automatically added to the face group.
7. Call `ISMNormalCutFeatureData2::CreateGroup` to access `ISMNormalCutGroupData`.
8. Repeat steps 6 and 7 as many times as you have face groups. Call `IModelDoc2::ClearSelection2` after each face group is created.
9. Call `IFeatureManager::CreateFeature`.

To edit a normal cut feature:

1. Call `IFeature::GetDefinition` to get this feature data object for a selected normal cut feature.
2. Call `ISMNormalCutFeatureData2::AccessSelections`. **Note: Any selections made *before* this call are invalid. To edit the properties of this feature, you must re-select their entities during the editing session, i.e., after calling `ISMNormalCutFeatureData2::AccessSelections`. (See step 5.)**
3. Call `ISMNormalCutFeatureData2::GetGroupNames` and `ISMNormalCutFeatureData2::GetGroupByName` to access `ISMNormalCutGroupData` for an existing face group.
4. Call `ISMNormalCutFeatureData2::CreateGroup` to create another face group if needed.
5. Modify [cut direction](#), [offset plane reference](#), or other properties on `ISMNormalCutFeatureData2` as needed. To select property entities, use `IModelDocExtension::SelectByRay` or `IModelDocExtension::SelectById2` followed by `ISelectionMgr::GetSelectedObject6`. You can also use `IEntity::Select4`.
6. Call `IFeature::ModifyDefinition` if you modified the feature or `ISMNormalCutFeatureData2::ReleaseSelectionAccess` if you did not.

Accessors

[IFeature::GetDefinition](#)

[IFeatureManager::CreateDefinition](#)

Access Diagram

[SMNormalCutFeatureData2](#)

See Also

[ISMNormalCutFeatureData2 Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

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ISMNormalCutFeatureData2 Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISMNormalCutFeatureData2 Interface

The following tables list the members exposed by [ISMNormalCutFeatureData2](#).

Public Properties

Name	Description
 AutoPropagation	Gets or sets whether to automatically add tangent faces to the cut-extrude face group.
 CutDirection	Gets or sets the edge, linear curve, or planar face that defines the direction of the Normal Cut.
 LayerAdjustmentValue	Gets or sets the offset plane adjustment value.
 LinkToKFactor	Gets or sets whether to link to a K-Factor when adjusting the offset plane of this Normal Cut.
 NormalCutParameters	Gets or sets the Normal Cut parameters.
 OffsetPlaneReference	Gets or sets the offset plane reference for the Normal Cut.
 OptimizeGeometry	Gets or sets whether to optimize the geometry of the Normal Cut.

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Public Methods

Name	Description
 AccessSelections	Gains access to the selections defining this sheet metal Normal Cut feature.
 CreateGroup	Creates a group of connected faces to cut.
 DeleteGroupByName	Deletes the specified face group.
 GetGroupByName	Gets the specified face group.
 GetGroupNames	Gets an array of face group names.
 ReleaseSelectionAccess	Releases access to the selections that define this sheet metal Normal Cut feature.

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See Also

[ISMNormalCutFeatureData2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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ISMNormalCutFeatureData2 Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISMNormalCutFeatureData2 Interface

For a list of all members of this type, see [ISMNormalCutFeatureData2 members](#).

Public Methods

Name	Description
 AccessSelections	Gains access to the selections defining this sheet metal Normal Cut feature.
 CreateGroup	Creates a group of connected faces to cut.
 DeleteGroupByName	Deletes the specified face group.
 GetGroupByName	Gets the specified face group.
 GetGroupNames	Gets an array of face group names.
 ReleaseSelectionAccess	Releases access to the selections that define this sheet metal Normal Cut feature.

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 See Also

[ISMNormalCutFeatureData2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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ISMNormalCutFeatureData2 Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISMNormalCutFeatureData2 Interface

For a list of all members of this type, see [ISMNormalCutFeatureData2 members](#).

 Public Properties

Name	Description
 AutoPropagation	Gets or sets whether to automatically add tangent faces to the cut-extrude face group.
 CutDirection	Gets or sets the edge, linear curve, or planar face that defines the direction of the Normal Cut.
 LayerAdjustmentValue	Gets or sets the offset plane adjustment value.
 LinkToKFactor	Gets or sets whether to link to a K-Factor when adjusting the offset plane of this Normal Cut.
 NormalCutParameters	Gets or sets the Normal Cut parameters.
 OffsetPlaneReference	Gets or sets the offset plane reference for the Normal Cut.
 OptimizeGeometry	Gets or sets whether to optimize the geometry of the Normal Cut.

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 See Also

[ISMNormalCutFeatureData2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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AccessSelections Method (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : AccessSelections Method (ISMNormalCutFeatureData2)

Gains access to the selections defining this sheet metal Normal Cut feature.

 .NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections( _  
    ByVal TopDoc As System.Object, _  
    ByVal Component As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ISMNormalCutFeatureData2  
Dim TopDoc As System.Object  
Dim Component As System.Object
```



```
Dim value As System.Boolean

value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(
    System.object TopDoc,
    System.object Component
)
```

C++/CLI

```
System.bool AccessSelections(
    & System.Object^ TopDoc,
    & System.Object^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc
Top-level document
Component
Component for the feature

Return Value

True if the selections were successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [ISMNormalCutFeatureData2::AccessSelections](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [ISMNormalCutFeatureData2::ReleaseSelectionAccess](#) if you did not

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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AutoPropagation Property (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : AutoPropagation Property (ISMNormalCutFeatureData2)

Gets or sets whether to automatically add tangent faces to the cut-extrude face group.

.NET Syntax

Visual Basic (Declaration)

Property AutoPropagation As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2
Dim value As System.Boolean

instance.AutoPropagation = value

value = instance.AutoPropagation
```

C#

```
System.bool AutoPropagation {get; set;}
```

C++/CLI

```
property System.bool AutoPropagation {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to add tangent faces, false to not

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::AutoPropagation](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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CreateGroup Method (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : CreateGroup Method (ISMNormalCutFeatureData2)

Creates a group of connected faces to cut.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateGroup( _
    ByRef ErrCode As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2
Dim ErrCode As System.Integer
Dim value As System.Object

value = instance.CreateGroup(ErrCode)
```

C#

```
System.Object CreateGroup(
    out System.int ErrCode
)
```

C++/CLI

```
System.Object^ CreateGroup(
    & [Out] System.int ErrCode
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ErrCode
Error code as defined in [swNormalCutErrors_e](#)

Return Value

[ISMNormalCutGroupData](#)

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::CreateGroup](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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CutDirection Property (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : CutDirection Property (ISMNormalCutFeatureData2)

Gets or sets the edge, linear curve, or planar face that defines the direction of the Normal Cut.

.NET Syntax

Visual Basic (Declaration)

```
Property CutDirection As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2
Dim value As System.Object

instance.CutDirection = value

value = instance.CutDirection
```

C#

```
System.Object CutDirection {get; set;}
```

C++/CLI

```
property System.Object^ CutDirection {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Edge](#), [curve](#), or [face](#) that defines the direction of the Normal Cut

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::CutDirection](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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DeleteGroupByName Method (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : DeleteGroupByName Method (ISMNormalCutFeatureData2)

Deletes the specified face group.

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteGroupByName( _  
    ByVal GroupName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2  
Dim GroupName As System.String  
Dim value As System.Boolean  
  
value = instance.DeleteGroupByName(GroupName)
```

C#

```
System.bool DeleteGroupByName(  
    System.string GroupName  
)
```

C++/CLI

```
System.bool DeleteGroupByName(  
& System.String^ GroupName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GroupName
Name of face group to delete

Return Value

True if deletion successful, false if not

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::DeleteGroupName](#).

Remarks

Use [ISMNormalCutFeatureData2::GetGroupNames](#) to populate GroupName.

You cannot delete a group, if it is the only group that exists.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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GetGroupName Method (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : GetGroupName Method (ISMNormalCutFeatureData2)

Gets the specified face group.

.NET Syntax

Visual Basic (Declaration)

```
Function GetGroupName( _  
    ByVal GroupName As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2  
Dim GroupName As System.String  
Dim value As System.Object  
  
value = instance.GetGroupName(GroupName)
```

C#

```
System.object GetGroupName(  
    System.string GroupName  
)
```

C++/CLI

```
System.Object^ GetGroupName(  
& System.String^ GroupName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GroupName
Name of face group to retrieve

Return Value

[ISMNormalCutGroupData](#)

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::GetGroupByName](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

Remarks

Use [ISMNormalCutFeatureData2::GetGroupNames](#) to populate GroupName.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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GetGroupNames Method (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : GetGroupNames Method (ISMNormalCutFeatureData2)

Gets an array of face group names.

.NET Syntax

Visual Basic (Declaration)

```
Function GetGroupNames() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2  
Dim value As System.Object
```

```
value = instance.GetGroupNames()
```

C#

```
System.object GetGroupNames()
```

C++/CLI

```
System.Object^ GetGroupNames();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of face group names

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::GetGroupNames](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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LayerAdjustmentValue Property (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : LayerAdjustmentValue Property (ISMNormalCutFeatureData2)

Gets or sets the offset plane adjustment value.

.NET Syntax

Visual Basic (Declaration)

```
Property LayerAdjustmentValue As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2  
Dim value As System.Double
```

```
instance.LayerAdjustmentValue = value
```

```
value = instance.LayerAdjustmentValue
```

C#

```
System.double LayerAdjustmentValue {get; set;}
```

C++/CLI

```
property System.double LayerAdjustmentValue {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

0.0 <= offset <= 1.0

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::LayerAdjustmentValue](#).

Remarks

This property is valid only if [ISMNormalCutFeatureData2::NormalCutParameters](#) is set to [swNormalCutParameters_e.swNormalCutOffsetPlane](#).
If [ISMNormalCutFeatureData2::LinkToKFactor](#) is set to true, this property is automatically set to 0.5.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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LinkToKFactor Property (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : LinkToKFactor Property (ISMNormalCutFeatureData2)

Gets or sets whether to link to a K-Factor when adjusting the offset plane of this Normal Cut.

.NET Syntax

Visual Basic (Declaration)

Property LinkToKFactor As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2  
Dim value As System.Boolean
```

```
instance.LinkToKFactor = value
```

```
value = instance.LinkToKFactor
```

C#

```
System.bool LinkToKFactor {get; set;}
```

C++/CLI

```
property System.bool LinkToKFactor {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to link to a K-Factor, false to not

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::LinkToKFactor](#).

Remarks

This property is valid only if:

- [ISMNormalCutFeatureData2::NormalCutParameters](#) is set to [swNormalCutParameters_e.swNormalCutOffsetPlane](#)
- and -
- [ISMNormalCutFeatureData2::OffsetPlaneReference](#) is set.

If this property is set to true, then [ISMNormalCutFeatureData2::LayerAdjustmentValue](#) is automatically set to 0.5.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

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NormalCutParameters Property (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : NormalCutParameters Property (ISMNormalCutFeatureData2)

Gets or sets the Normal Cut parameters.

.NET Syntax

```
Property NormalCutParameters As System.Integer
```



```
Dim instance As ISMNormalCutFeatureData2
Dim value As System.Integer

instance.NormalCutParameters = value

value = instance.NormalCutParameters
```

```
System.int NormalCutParameters {get; set;}
```

```
property System.int NormalCutParameters {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Normal Cut parameters as defined in [swNormalCutParameters_e](#)

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::NormalCutParameters](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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OffsetPlaneReference Property (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : OffsetPlaneReference Property (ISMNormalCutFeatureData2)

Gets or sets the offset plane reference for the Normal Cut.

.NET Syntax

```
Property OffsetPlaneReference As System.Object
```



```
Dim instance As ISMNormalCutFeatureData2
Dim value As System.Object

instance.OffsetPlaneReference = value

value = instance.OffsetPlaneReference
```

C#

```
System.Object OffsetPlaneReference {get; set;}
```

C++/CLI

```
property System.Object^ OffsetPlaneReference {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Reference [plane](#) or top or bottom [face](#) of the sheet metal body

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::OffsetPlaneReference](#).

Remarks

This property is valid only if [ISMNormalCutFeatureData2::NormalCutParameters](#) is set to [swNormalCutParameters_e.swNormalCutOffsetPlane](#).

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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OptimizeGeometry Property (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : OptimizeGeometry Property (ISMNormalCutFeatureData2)

Gets or sets whether to optimize the geometry of the Normal Cut.

.NET Syntax

Visual Basic (Declaration)

```
Property OptimizeGeometry As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2  
Dim value As System.Boolean
```

```
instance.OptimizeGeometry = value
```

```
value = instance.OptimizeGeometry
```

C#

```
System.bool OptimizeGeometry {get; set;}
```

C++/CLI

```
property System.bool OptimizeGeometry {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to optimize the geometry of the Normal Cut, false to not

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::OptimizeGeometry](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

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ReleaseSelectionAccess Method (ISMNormalCutFeatureData2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutFeatureData2 Interface](#) : ReleaseSelectionAccess Method (ISMNormalCutFeatureData2)

Releases access to the selections that define this sheet metal Normal Cut feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)



```
Dim instance As ISMNormalCutFeatureData2
```

```
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutFeatureData2::ReleaseSelectionAccess](#).

Remarks

[ISMNormalCutFeatureData2::AccessSelections](#) puts the model in a rollback state to allow access to the selections that define this feature.

Use this method to restore the rollback state if you did not modify the feature, or use [IFeature::ModifyDefinition](#) if you did modify the feature.

See Also

[ISMNormalCutFeatureData2 Interface](#)
[ISMNormalCutFeatureData2 Members](#)

Availability

ISMNormalCutGroupData (Type)

ISMNormalCutGroupData Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISMNormalCutGroupData Interface

Allows access to a sheet metal normal cut feature's cut-extrude face group.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISMNormalCutGroupData
```

Visual Basic (Usage)

```
Dim instance As ISMNormalCutGroupData
```

C#

```
public interface ISMNormalCutGroupData
```

C++/CLI

```
public interface class ISMNormalCutGroupData
```

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutGroupData](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

Remarks

See the [ISMNormalCutFeatureData2](#) Remarks.

Accessors

[ISMNormalCutFeatureData2::CreateGroup](#)

[ISMNormalCutFeatureData2::GetGroupByName](#)

Access Diagram

[SMNormalCutGroupData](#)

See Also

[ISMNormalCutGroupData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISMNormalCutGroupData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISMNormalCutGroupData Interface

For a list of all members of this type, see [ISMNormalCutGroupData members](#).

Public Methods

Name	Description
 GetGroupName	Gets the name of the group.

[Top](#)

See Also

[ISMNormalCutGroupData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISMNormalCutGroupData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISMNormalCutGroupData Interface

The following tables list the members exposed by [ISMNormalCutGroupData](#).

Public Properties

Name	Description
 Faces	Gets or sets the cut-extrude faces in this group.

[Top](#)

Public Methods

Name	Description
 GetGroupName	Gets the name of the group.

[Top](#)

See Also

[ISMNormalCutGroupData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISMNormalCutGroupData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISMNormalCutGroupData Interface

For a list of all members of this type, see [ISMNormalCutGroupData members](#).

Public Properties

Name	Description
 Faces	Gets or sets the cut-extrude faces in this group.

[Top](#)

See Also

[ISMNormalCutGroupData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Faces Property (ISMNormalCutGroupData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutGroupData Interface](#) : Faces Property (ISMNormalCutGroupData)

Gets or sets the cut-extrude faces in this group.

.NET Syntax

Visual Basic (Declaration)

Property Faces As System.Object

```
Dim instance As ISMNormalCutGroupData
Dim value As System.Object

instance.Faces = value

value = instance.Faces
```

C#

```
System.Object Faces {get; set;}
```

C++/CLI

```
property System.Object^ Faces {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [IFace2](#)

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutGroupData::Faces](#).

Example

See the [ISMNormalCutFeatureData2](#) example.

See Also

[ISMNormalCutGroupData Interface](#)
[ISMNormalCutGroupData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

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GetGroupName Method (ISMNormalCutGroupData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISMNormalCutGroupData Interface](#) : GetGroupName Method (ISMNormalCutGroupData)

Gets the name of the group.

.NET Syntax

Visual Basic (Declaration)

```
Function GetGroupName() As System.String
```

Visual Basic (Usage)

```
Dim instance As ISMNormalCutGroupData
Dim value As System.String

value = instance.GetGroupName()
```

C#

```
System.string GetGroupName()
```

C++/CLI

```
System.String^ GetGroupName();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Name of the group

Visual Basic for Applications (VBA) Syntax

See [SMNormalCutGroupData::GetGroupName](#).

See Also

[ISMNormalCutGroupData Interface](#)
[ISMNormalCutGroupData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

ISnapShot (Type)

ISnapShot Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISnapShot Interface

Allows access to the snapshot of the graphics area of an assembly opened in Large Design Review mode.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface ISnapShot

Visual Basic (Usage)



Dim instance As [ISnapShot](#)

C#

public interface ISnapShot

C++/CLI

public interface class ISnapShot

Visual Basic for Applications (VBA) Syntax

See [SnapShot](#).

Example

[Open Assembly in Large Design Review Mode \(VBA\)](#)
[Open Assembly in Large Design Review Mode \(VB.NET\)](#)
[Open Assembly in Large Design Review Mode \(C#\)](#)

Remarks

To open an assembly in Large Design Review mode:

1. Call [ISldWorks::GetOpenDocSpec](#) to create an instance of [IDocumentSpecification](#).
2. Set [IDocumentSpecification::ViewOnly](#) to true.
3. Call [ISldWorks::OpenDoc7](#), passing it the instance of IDocumentSpecification.

- or -

Call [ISldWorks::OpenDoc6](#) with the Options parameter set to swOpenDocOptions_e.swOpenDocOptions_ViewOnly.

Accessors

[IModelViewManager::AddSnapShot](#)

[IModelViewManager::GetSnapShot](#)

[IModelViewManager::GetSnapShots](#)

Access Diagram

[SnapShot](#)

See Also

[ISnapShot Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISnapShot Interface Members

[See Also](#) [Properties](#) [Methods](#)


SolidWorks.Interop.sldworks Namespace : ISnapShot Interface

The following tables list the members exposed by [ISnapShot](#).

 **Public Properties**

Name	Description
 Comment	Gets or sets the comment on this snapshot.
 Name	Gets or sets the name of this snapshot.
 ViewId	Gets the ID of this snapshot.

[Top](#)

 **Public Methods**

Name	Description
 Activate	Activates this snapshot.

[Top](#)

 **See Also**

[ISnapShot Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISnapShot Interface Methods

[See Also](#)


SolidWorks.Interop.sldworks Namespace : ISnapShot Interface

For a list of all members of this type, see [ISnapShot members](#).

 **Public Methods**

Name	Description
 Activate	Activates this snapshot.

[Top](#)

 **See Also**

[ISnapShot Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISnapShot Interface Properties

[See Also](#)


SolidWorks.Interop.sldworks Namespace : ISnapShot Interface

For a list of all members of this type, see [ISnapShot members](#).

 **Public Properties**

Name	Description
 Comment	Gets or sets the comment on this snapshot.
 Name	Gets or sets the name of this snapshot.
 ViewId	Gets the ID of this snapshot.

[Top](#)

 **See Also**

Activate Method (ISnapShot)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISnapShot Interface](#) : Activate Method (ISnapShot)

Activates this snapshot.

.NET Syntax

Visual Basic (Declaration)

```
Function Activate() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISnapShot  
Dim value As System.Boolean  
  
value = instance.Activate()
```

C#

```
System.bool Activate()
```

C++/CLI

```
System.bool Activate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [SnapShot::Activate](#).

Example

[Open Assembly in Large Design Review Mode \(C#\)](#)
[Open Assembly in Large Design Review Mode \(VB.NET\)](#)
[Open Assembly in Large Design Review Mode \(VBA\)](#)

See Also

[ISnapShot Interface](#)
[ISnapShot Members](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

Comment Property (ISnapShot)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISnapShot Interface](#) : Comment Property (ISnapShot)

Gets or sets the comment on this snapshot.

.NET Syntax

Visual Basic (Declaration)

```
Property Comment As System.String
```

```
Dim instance As ISnapShot
Dim value As System.String

instance.Comment = value

value = instance.Comment
```

C#

```
System.string Comment {get; set;}
```

C++/CLI

```
property System.String^ Comment {
    System.String^ get();
    void set ( & System.String^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Comment (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [SnapShot.Comment](#).

Example

[Open Assembly in Large Design Review Mode \(VBA\)](#)
[Open Assembly in Large Design Review Mode \(VB.NET\)](#)
[Open Assembly in Large Design Review Mode \(C#\)](#)

Remarks

To add a time stamp to the comment, prepend the comment string with "<TS>".

For example:

```
snap1.Comment = "<TS> This is a comment for SnapShot1"
```

generates:

```
07/29/2011 04:08 PM user_id: This is a comment for SnapShot1
```

See Also

[ISnapShot Interface](#)
[ISnapShot Members](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

Name Property (ISnapShot)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISnapShot Interface](#) : Name Property (ISnapShot)

Gets or sets the name of this snapshot.

.NET Syntax

Visual Basic (Declaration)

```
Property Name As System.String
```

```
Dim instance As ISnapShot
Dim value As System.String

instance.Name = value

value = instance.Name
```

C#

```
System.string Name {get; set;}
```

C++/CLI

```
property System.String^ Name {
    System.String^ get();
    void set ( & System.String^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

- Name of the snapshot
- OR -
- "" to give a default name of "Snapn"
- OR -
- "?" to open the Name Snapshot dialog box

Visual Basic for Applications (VBA) Syntax

See [SnapShot::Name](#).

Example

[Open Assembly in Large Design Review Mode \(C#\)](#)
[Open Assembly in Large Design Review Mode \(VB.NET\)](#)
[Open Assembly in Large Design Review Mode \(VBA\)](#)

See Also

[ISnapShot Interface](#)
[ISnapShot Members](#)

Availability

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ViewId Property (ISnapShot)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISnapShot Interface](#) : ViewId Property (ISnapShot)

Gets the ID of this snapshot.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ViewId As System.Integer
```

Visual Basic (Usage)

Copy Code

```
Dim instance As ISnapShot
Dim value As System.Integer

value = instance.ViewId
```

C#

```
System.int ViewId {get;}
```

C++/CLI

```
property System.int ViewId {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Snapshot ID

Visual Basic for Applications (VBA) Syntax

See [SnapShot::ViewId](#).

See Also

[ISnapShot Interface](#)
[ISnapShot Members](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

ISplineHandle (Type)

ISplineHandle Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISplineHandle Interface

Provides access to spline handles.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISplineHandle
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle
```

C#

```
public interface ISplineHandle
```

C++/CLI

```
public interface class ISplineHandle
```

Visual Basic for Applications (VBA) Syntax

See [SplineHandle](#).

Example

[Get and Set Spline Handles \(VBA\)](#)
[Get and Set Spline Handles \(VB.NET\)](#)
[Get and Set Spline Handles \(C#\)](#)

Remarks

To enable spline tangency and curvature handles, set [ISldWorks::SetUserPreferenceToggle](#) swDisplayAllSplineHandles to True.

NOTE: [ISelectionMgr::GetSelectedObjectType3](#) returns swSelMANIPULATORS for a spline handle.

Accessors

[ISketchSpline::AddCurvatureControl](#)
[ISketchSpline::AddTangencyControl](#)
[ISketchSpline::GetSplineHandles](#) and [ISketchSpline::!GetSplineHandles](#)

Access Diagram

[SplineHandle](#)

See Also

[ISplineHandle Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchSpline Interface](#)

ISplineHandle Interface Members

[See Also](#) [Properties](#) [Methods](#)



The following tables list the members exposed by [ISplineHandle](#).

 Public Properties

Name	Description
 Curvature	Gets or sets the degree of curvature at any point where curvature control was added.
 CurvatureControlled	Gets or sets whether the spline handle has curvature control.
 Editable	Gets or sets whether this spline handle can be edited.
 Parameter	Gets value in the parameter space of the curve underlying the spline handle.
 RadiusOfCurvature	Gets or sets the radius of curvature at any spline point.
 SplinePointNumber	Gets the number of the points of this spline handle.
 TangentDriving	Enables or disables spline control using tangent magnitude, tangent radial direction , and tangent polar direction .
 TangentMagnitude	Gets or sets the weight for the tangency magnitude in the specified direction for this spline handle.
 TangentPolarDirection	Gets or sets the tangent polar direction, which is the elevation angle of the tangent vector to a plane placed at a point perpendicular to a spline point.
 TangentRadialDirection	Gets or sets the tangent radial direction, which controls the tangency direction by allowing modification of the spline's angle of inclination relative to the x, y, or z axis.
 X	Gets or sets the x coordinate of the spline point.
 Y	Gets or sets the y coordinate of the spline point.
 Z	Gets or sets the z coordinate of the spline point.

[Top](#)

 Public Methods

Name	Description
 Reset	Resets the selected handle to its initial state on this spline handle.
 Select	Selects a spline handle.

[Top](#)

 See Also

[ISplineHandle Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchSpline Interface](#)

ISplineHandle Interface Methods

[See Also](#)



For a list of all members of this type, see [ISplineHandle members](#).

 Public Methods

Name	Description
 Reset	Resets the selected handle to its initial state on this spline handle.
 Select	Selects a spline handle.

[Top](#)

 See Also

[ISplineHandle Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchSpline Interface](#)

ISplineHandle Interface Properties

[See Also](#)



For a list of all members of this type, see [ISplineHandle members](#).

Public Properties

Name	Description
Curvature	Gets or sets the degree of curvature at any point where curvature control was added.
CurvatureControlled	Gets or sets whether the spline handle has curvature control.
Editable	Gets or sets whether this spline handle can be edited.
Parameter	Gets value in the parameter space of the curve underlying the spline handle.
RadiusOfCurvature	Gets or sets the radius of curvature at any spline point.
SplinePointNumber	Gets the number of the points of this spline handle.
TangentDriving	Enables or disables spline control using tangent magnitude, tangent radial direction , and tangent polar direction .
TangentMagnitude	Gets or sets the weight for the tangency magnitude in the specified direction for this spline handle.
TangentPolarDirection	Gets or sets the tangent polar direction, which is the elevation angle of the tangent vector to a plane placed at a point perpendicular to a spline point.
TangentRadialDirection	Gets or sets the tangent radial direction, which controls the tangency direction by allowing modification of the spline's angle of inclination relative to the x, y, or z axis.
X	Gets or sets the x coordinate of the spline point.
Y	Gets or sets the y coordinate of the spline point.
Z	Gets or sets the z coordinate of the spline point.

[Top](#)

See Also

[ISplineHandle Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchSpline Interface](#)

Curvature Property (ISplineHandle)

[See Also](#)



Gets or sets the degree of curvature at any point where curvature control was added.

.NET Syntax

Visual Basic (Declaration)

Property Curvature As System.Double

Visual Basic (Usage)



```
Dim instance As ISplineHandle  
Dim value As System.Double
```

```
instance.Curvature = value
```

```
value = instance.Curvature
```

C#

```
System.double Curvature {get; set;}
```

C++/CLI

```
property System.double Curvature {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Degree of curvature

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::Curvature](#).

Example

See the [ISplineHandle](#) examples.

Remarks

Curvature control must be enabled for this property to have an effect. Use [ISplineHandle::CurvatureControlled](#) to see if curvature control is enabled or to set curvature control for this spline handle.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle::RadiusOfCurvature Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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CurvatureControlled Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : CurvatureControlled Property (ISplineHandle)

Gets or sets whether the spline handle has curvature control.

.NET Syntax

Visual Basic (Declaration)

```
Property CurvatureControlled As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle
Dim value As System.Boolean

instance.CurvatureControlled = value
value = instance.CurvatureControlled
```

C#

```
System.bool CurvatureControlled {get; set;}
```

C++/CLI

```
property System.bool CurvatureControlled {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the spline handle has curvature control, false if not

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::CurvatureControlled](#).

Remarks

If the spline handle has curvature control, you can use:

- [ISplineHandle::Curvature](#)
- [ISplineHandle::RadiusOfCurvature](#)

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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Editable Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : Editable Property (ISplineHandle)

Gets or sets whether this spline handle can be edited.

.NET Syntax

Visual Basic (Declaration)

Property Editable As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISplineHandle  
Dim value As System.Boolean
```

```
instance.Editable = value
```

```
value = instance.Editable
```

C#

```
System.bool Editable {get; set;}
```

C++/CLI

```
property System.bool Editable {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the spline can be edited, false if not

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::Editable](#).

See Also

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Parameter Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : Parameter Property (ISplineHandle)

Gets value in the parameter space of the curve underlying the spline handle.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Parameter As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle  
Dim value As System.Double  
  
value = instance.Parameter
```

C#

```
System.double Parameter {get;}
```

C++/CLI

```
property System.double Parameter {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Value in the parameter space of the curve underlying the spline handle

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::Parameter](#).

Remarks

There is no user-interface equivalent for this property as it is typically only useful for API client code.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RadiusOfCurvature Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : RadiusOfCurvature Property (ISplineHandle)

Gets or sets the radius of curvature at any spline point.

.NET Syntax

Visual Basic (Declaration)

```
Property RadiusOfCurvature As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle  
Dim value As System.Double  
  
instance.RadiusOfCurvature = value  
value = instance.RadiusOfCurvature
```

C#

```
System.double RadiusOfCurvature {get; set;}
```

C++/CLI

```
property System.double RadiusOfCurvature {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Radius of curvature

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::RadiusOfCurvature](#).

Example

See the [ISplineHandle](#) examples.

Remarks

Curvature control must be enabled for this property to have an effect. Use [ISplineHandle::CurvatureControlled](#) to see if curvature control is enabled or to set curvature control for this spline handle.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle::Curvature Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Reset Method (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : Reset Method (ISplineHandle)

Resets the selected handle to its initial state on this spline handle.

.NET Syntax

Visual Basic (Declaration)

```
Sub Reset()
```

```
Dim instance As ISplineHandle

instance.Reset()
```

C#

```
void Reset()
```

C++/CLI

```
void Reset();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::Reset](#).

Example

See the [ISplineHandle](#) examples.

Remarks

Use [ISketchSpline::ResetAllHandles](#) to reset all spline handles to their initial state.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Select Method (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : Select Method (ISplineHandle)

Selects a spline handle.

.NET Syntax

Visual Basic (Declaration)

```
Function Select( _
    ByVal AppendFlag As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)

Copy Code

```
Dim instance As ISplineHandle
Dim AppendFlag As System.Boolean
Dim value As System.Boolean

value = instance.Select(AppendFlag)
```

C#

```
System.bool Select(
    System.bool AppendFlag
)
```

```
System.bool Select(
& System.bool AppendFlag
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppendFlag

True appends the spline handle to the selection list, false replaces the selection list with the spline handle

Return Value

True if the spline handle is selected, false if not

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::Select](#).

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SplinePointNumber Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : SplinePointNumber Property (ISplineHandle)

Gets the number of the points of this spline handle.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property SplinePointNumber As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle
Dim value As System.Integer

value = instance.SplinePointNumber
```

C#

```
System.int SplinePointNumber {get;}
```

C++/CLI

```
property System.int SplinePointNumber {
    System.int get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of the point of this handle

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::SplinePointNumber](#).

Example

See the [ISplineHandle](#) examples.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISketchSpline.DeletePoint Method](#)
[ISketchSpline.GetPoints2 Method](#)
[ISketchSpline.InsertPoint Method](#)
[ISketchSpline.EnumPoints Method](#)

Availability

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TangentDriving Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : TangentDriving Property (ISplineHandle)

Enables or disables spline control using tangent magnitude, [tangent radial direction](#), and [tangent polar direction](#).

.NET Syntax

Visual Basic (Declaration)

```
Property TangentDriving As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle  
Dim value As System.Boolean
```

```
instance.TangentDriving = value
```

```
value = instance.TangentDriving
```

C#

```
System.bool TangentDriving {get; set;}
```

C++/CLI

```
property System.bool TangentDriving {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to enable spline control using tangent magnitude and tangent radial, false to not

Visual Basic for Applications (VBA) Syntax

See [SplineHandle.TangentDriving](#).

Example

See the [ISplineHandle](#) examples.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle.TangentMagnitude Property](#)

[ISplineHandle::TangentPolarDirection Property](#)
[ISplineHandle::TangentRadialDirection Property](#)

Availability

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TangentMagnitude Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : TangentMagnitude Property (ISplineHandle)

Gets or sets the weight for the tangency magnitude in the specified direction for this spline handle.

.NET Syntax

Visual Basic (Declaration)

```
Property TangentMagnitude( _  
    ByVal WhichDirection As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle  
Dim WhichDirection As System.Integer  
Dim value As System.Double  
  
instance.TangentMagnitude(WhichDirection) = value  
  
value = instance.TangentMagnitude(WhichDirection)
```

C#

```
System.double TangentMagnitude(  
    System.int WhichDirection  
) {get; set;}
```

C++/CLI

```
property System.double TangentMagnitude {  
    System.double get(System.int WhichDirection);  
    void set (System.int WhichDirection, System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichDirection
Direction as defined in [swTangentMagnitudeDirection_e](#)

Property Value

Weight for the tangency magnitude in the specified direction

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::TangentMagnitude](#).

Example

See the [ISplineHandle](#) examples.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle::TangentDriving Property](#)
[ISplineHandle::TangentPolarDirection Property](#)
[ISplineHandle::TangentRadialDirection Property](#)

Availability

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TangentPolarDirection Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : TangentPolarDirection Property (ISplineHandle)

Gets or sets the tangent polar direction, which is the elevation angle of the tangent vector to a plane placed at a point perpendicular to a spline point.

.NET Syntax

Visual Basic (Declaration)

Property TangentPolarDirection As System.Double

Visual Basic (Usage)



```
Dim instance As ISplineHandle
Dim value As System.Double

instance.TangentPolarDirection = value
value = instance.TangentPolarDirection
```

C#

```
System.double TangentPolarDirection {get; set;}
```

C++/CLI

```
property System.double TangentPolarDirection {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Tangent polar direction

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::TangentPolarDirection](#).

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle::TangentDriving Property](#)
[ISplineHandle::TangentMagnitude Property](#)
[ISplineHandle::TangentRadialDirection Property](#)

Availability

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TangentRadialDirection Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : TangentRadialDirection Property (ISplineHandle)

Gets or sets the tangent radial direction, which controls the tangency direction by allowing modification of the spline's angle of inclination relative to the x, y, or z axis.

.NET Syntax

Visual Basic (Declaration)

Property TangentRadialDirection As System.Double

Visual Basic (Usage)



```
Dim instance As ISplineHandle
Dim value As System.Double

instance.TangentRadialDirection = value

value = instance.TangentRadialDirection
```

C#

```
System.double TangentRadialDirection {get; set;}
```

C++/CLI

```
property System.double TangentRadialDirection {
    System.double get();
    void set ( &    System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Tangent radial direction

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::TangentRadialDirection](#).

Example

See the [ISplineHandle](#) examples.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle::TangentDriving Property](#)
[ISplineHandle::TangentMagnitude Property](#)
[ISplineHandle::TangentPolarDirection Property](#)

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X Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : X Property (ISplineHandle)

Gets or sets the x coordinate of the spline point.

.NET Syntax

Visual Basic (Declaration)

```
Property X As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle
Dim value As System.Double

instance.X = value
```

```
value = instance.X
```

C#

```
System.double X {get; set;}
```

C++/CLI

```
property System.double X {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

x coordinate of the spline point

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::X](#).

Example

See the [ISplineHandle](#) examples.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle::Y Property](#)
[ISplineHandle::Z Property](#)

Availability

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Y Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : Y Property (ISplineHandle)

Gets or sets the y coordinate of the spline point.

.NET Syntax

Visual Basic (Declaration)

```
Property Y As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISplineHandle  
Dim value As System.Double
```

```
instance.Y = value
```

```
value = instance.Y
```

C#

```
System.double Y {get; set;}
```

C++/CLI

```
property System.double Y {  
    System.double get();
```

```
void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

y coordinate of the spline point

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::Y](#).

Example

See the [ISplineHandle](#) examples.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle::X Property](#)
[ISplineHandle::Z Property](#)

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Z Property (ISplineHandle)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISplineHandle Interface](#) : Z Property (ISplineHandle)

Gets or sets the z coordinate of the spline point.

.NET Syntax

Visual Basic (Declaration)

Property Z As System.Double

Visual Basic (Usage)



```
Dim instance As ISplineHandle  
Dim value As System.Double
```

```
instance.Z = value
```

```
value = instance.Z
```

C#

```
System.double Z {get; set;}
```

C++/CLI

```
property System.double Z {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

z coordinate of the spline point

Visual Basic for Applications (VBA) Syntax

See [SplineHandle::Z](#).

Example

See the [ISplineHandle](#) examples.

See Also

[ISplineHandle Interface](#)
[ISplineHandle Members](#)
[ISplineHandle::X Property](#)
[ISplineHandle::Y Property](#)

Availability

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